

Galois 2

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Problem 1. *The galois group of a polynomial means the galois group of a splitting field. Explain why the galois group is a subgroup of $\text{Perm}(\text{roots of } f)$. Explain why f is irreducible iff this galois group acts transitively on the roots*

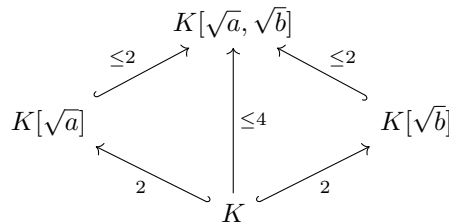
Proof. As previously shown, $\text{Gal}(K[f]/K)$ is in bijection with the roots of f with the evaluation homomorphism. Suppose r is a root of f , $\forall g \in \text{Gal}(K[f]/K)$, $f(g(r)) = g(f(r)) = 0$. Thus $g(r)$ is root. Thus g takes roots to roots. For 2 distinct roots since g is a automorphism it must take 2 distinct roots to 2 distinct roots thus, g is a permutation of roots. Furthermore since each element of Gal is determined by where it sends any root of f , no two elements of Gal cause the same permutations of roots thus $\text{Gal}(K[f]/K) \subseteq \text{Perm}(\text{roots of } f)$.

Suppose f is reducible. Thus $\exists g, h \in K[X] \ni f(x) = g(x)h(x)$ where wlog $g, h \notin K$ and $\gcd(g, h) = 1$. WLOG suppose r is a root of g and s is a root of h . Then $\forall \sigma \in \text{Gal}(K[f]/K)$, $\sigma(r)$ is a root of g and $\sigma(s)$ is a root of h . Thus there are 2 orbits, $\text{Gal}(K[f]/K) \cdot r$ and $\text{Gal}(K[f]/K) \cdot s$. Thus the action is not transitive. Thus if the action is transitive then its irreducible.

Suppose f is irreducible. Assume the contrary that G is not transitive. Then let s, r be elements of the roots of f in distinct orbits of G . Then let $g(x) = \prod_{\alpha \in H \cdot r} (x - \alpha)$ and $h(x) = \prod_{\beta \in H \cdot s} (x - \beta)$ and $g, h | f$. Moreover g, h are fixed by Gal and thus, $g, h \in K[x]$. Moreover by a previous homework, g, h are irreducible. Thus f factors into 2 irreducible polynomials thus f is reducible and thus by contradiction, G must be transitive. $\square$

Problem 2. *Consider $K[\sqrt{a}, \sqrt{b}]$ where a, b are in K and neither a nor b is a square. Char 0 (all you really need is char other than 2). Show the galois group is either $\mu_2 \times \mu_2$, or μ_2 , with μ_2 iff ab is a square. (μ_n means the multiplicative cyclic group).*

Proof. The minimum polynomial of $\sqrt{a}$ over K is $p_{\sqrt{a}}(x) = x^2 - a$. The minimum polynomial of $\sqrt{b}$ over K is $p_{\sqrt{b}}(x) = x^2 - b$. The roots to $p_{\sqrt{a}}(x) = x^2 - a$ are $\sqrt{a}$ and $-\sqrt{a}$. Similarly the roots to $p_{\sqrt{b}}(x) = x^2 - b$ are $\sqrt{b}$ and $-\sqrt{b}$. Thus, $K[\sqrt{a}]/K$ is galois and $K[\sqrt{a}, \sqrt{b}]/K$ is galois and $K[\sqrt{b}]/K$ is galois.



$\implies$: Suppose $ab \in K$ is a square. Then $\exists c^2 \in K \ni ab = c^2 \implies b = \frac{a}{c^2} \implies \sqrt{b} = \frac{\sqrt{a}}{c} \implies \sqrt{b} \in K[\sqrt{a}]$. Thus $K[\sqrt{a}, \sqrt{b}] = K[\sqrt{a}] \implies \text{Gal}(K[\sqrt{a}, \sqrt{b}]/K) = \text{Gal}(K[\sqrt{a}]/K)$. Since p_a is separable and splits over $K[\sqrt{a}] \implies \text{Gal}(K[\sqrt{a}]/K) = [K[\sqrt{a}] : K] = 2 \implies \text{Gal}(K[\sqrt{a}, \sqrt{b}]/K) = \mu_2$. Thus $\text{Gal}(K[\sqrt{a}, \sqrt{b}]/K) = \mu_2$.

$\Leftarrow$: Suppose $\text{Gal}(K[\sqrt{a}, \sqrt{b}]/K) = \mu_2$. Since $p_{\sqrt{a}}$ and $p_{\sqrt{b}}$ are separable and split over $K[\sqrt{a}, \sqrt{b}]$. Thus $K[\sqrt{a}, \sqrt{b}]/K$ is Galois. Thus, $[K[\sqrt{a}, \sqrt{b}] : K] = |\text{Gal}(K[\sqrt{a}, \sqrt{b}]/K)| = 2 \implies [K[\sqrt{a}, \sqrt{b}] : K[\sqrt{a}]] = 1 \implies \sqrt{b} \in K[\sqrt{a}]$. Thus $\exists x, y \in K \ni \sqrt{b} = x + y\sqrt{a} \implies b = x^2 + 2xy\sqrt{a} + ay^2$. Since $b \in K$ and $\sqrt{a} \notin K \implies xy = 0$. Assume the contrary, $y = 0$. Then $b = x^2$ and thus b is square in K which is a contradiction. Thus, $y \neq 0$ and $x = 0$. Thus $b = ay^2 \implies ab = a^2y^2 \implies \sqrt{ab} = \sqrt{a^2y^2} = ay \in K$. Thus ab is square in K .

Suppose ab is square in K . $\exists c^2 \in K \ni ab = c^2 \implies b = \frac{a}{c^2} \implies \sqrt{b} = \frac{\sqrt{a}}{c} \implies \sqrt{b} \in K[\sqrt{a}]$.

Suppose ab is not square in K . Thus $b \notin K[\sqrt{a}]$. Thus, $[K[\sqrt{a}, \sqrt{b}] : K[\sqrt{a}]] = 2 \implies [K[\sqrt{a}, \sqrt{b}] : K] = 4$. Since, $[K[\sqrt{a}, \sqrt{b}] : K] = 4$. Thus $\text{Gal}(K[\sqrt{a}, \sqrt{b}] : K) = \mu_4$ or $\text{Gal}(K[\sqrt{a}, \sqrt{b}] : K) = \mu_2 \times \mu_2$. By Galois Correspondence, since $[K[\sqrt{a}, \sqrt{b}] : K[\sqrt{a}]] = 2$ and $[K[\sqrt{a}, \sqrt{b}] : K[\sqrt{b}]] = 2$, and both extensions are Galois, $\text{Gal}(K[\sqrt{a}, \sqrt{b}] : K)$ has two subgroups isomorphic to μ_2 . Thus $\text{Gal}(K[\sqrt{a}, \sqrt{b}] : K) = \mu_2 \times \mu_2$. $\square$

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Problem 3. Consider a nested square root $a = \sqrt{r + \sqrt{t}}$ with $r, t \in F$. Assume α has degree 4 over F , let f be the irreducible polynomial of α over F , and let K be a splitting field of f over F .

Problem 3a. Compute the irreducible polynomial of α over F . Prove that $G(K/F)$ is one of the groups D_4 , C_4 , or D_2 .

Proof.

$$\begin{aligned}\alpha^4 &= (r + \sqrt{t})^2 \\ &= r^2 + 2r\sqrt{t} + t\end{aligned}$$

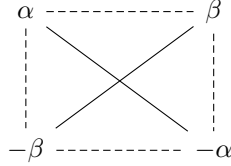
$$\begin{aligned}\alpha^4 - 2r\alpha^2 - t + r^2 &= r^2 + 2r\sqrt{t} + t - 2r(r + \sqrt{t}) - t + r^2 \\ &= r^2 - 2r^2 + r^2 + 2r\sqrt{t} - 2r\sqrt{t} + t - t \\ &= 0.\end{aligned}$$

Thus α is a solution to

$$x^4 - 2rx - t + r^2$$

Since α has degree 4, this is the minimum polynomial thus $f(x) = x^4 - 2rx + r^2 - t = (x^2 - r^2)^2 - t$. Let $\beta = \sqrt{r - \sqrt{t}}$. Thus the roots of f are $\pm\alpha, \pm\beta$. Let $S = \{\alpha, -\alpha, \beta, -\beta\}$. Thus $K = F[\alpha, \beta]$. β is a solution to $x^2 + \alpha^2 + r$ over $F[\alpha]$. Thus $[F[\alpha, \beta] : F[\alpha]] \leq 2$. Thus $[F[\alpha, \beta] : F] \leq 8$. Thus $|G(K/F)| \leq 8$.

Note for all $g \in \text{Gal}(K/F)$, $\forall s \in S$ if $g(-s) = -g(s)$. Thus a notion of "opposites" must be maintained. Thus the elements of S can be placed on the square any action of G must be a rigid transformation of the square which maintains the structure of opposite pairs.



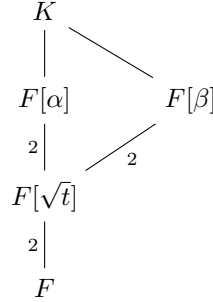
Thus $G \subseteq D_4 = \langle x, y | x^4 = y^2 = 1, yxy = x^{-1} \rangle$. Moreover G acts transitively on the roots by problem 1. The only subgroups of D_2 , $\langle x \rangle \cong \mu_4$, and $\langle x^2, y \rangle \cong K_4$, $\langle x^2 \rangle$, and $\langle y \rangle$.

D_4 is transitive. The orbit of the "vertex" α under μ_4 (not really the actual value), is $\alpha, -\beta, -\alpha, \beta$. The orbit of α over D_2 is $\alpha, -\beta, \beta, -\alpha$, and then cycles. Thus these three subgroups are transitive.

The orbit of α over $\langle x^2 \rangle$ is $\alpha, -\alpha, \alpha, \dots$. The orbit of α over $\langle y \rangle$ is $\alpha, \beta, \alpha, \dots$.

Thus the transitive subgroups of D_4 are D_4 , $\langle x \rangle \cong \mu_4$, and $\langle x^2, y \rangle \cong D_2$. $D_4, \mu_4 \triangleleft D_4$.

α and β are not squares over $F[\sqrt{t}]$.



$\alpha^2\beta^2 = r^2 - t$. By the previous problem, $r^2 - t$ not square over $F[\sqrt{t}] \iff \text{Gal}(K/F[\sqrt{t}]) = \mu_2 \times \mu_2 \iff [K : F[\sqrt{t}]] = 4 \iff [K : F] = 8 \iff \text{Gal}(K/F) = D_4$. Furthermore, $r^2 - t$ not square over $F[\sqrt{t}] \implies r^2 - t$ not square over F . Thus $r^2 - t$ not square over $F \implies \text{Gal}(K/F) = D_4$.

We have two generators of D_2 , σ, ρ where $\sigma(\alpha) = -\alpha$ and $\sigma(\beta) = -\beta$ and $\rho(\alpha) = \beta$ and $\rho(\beta) = \alpha$.

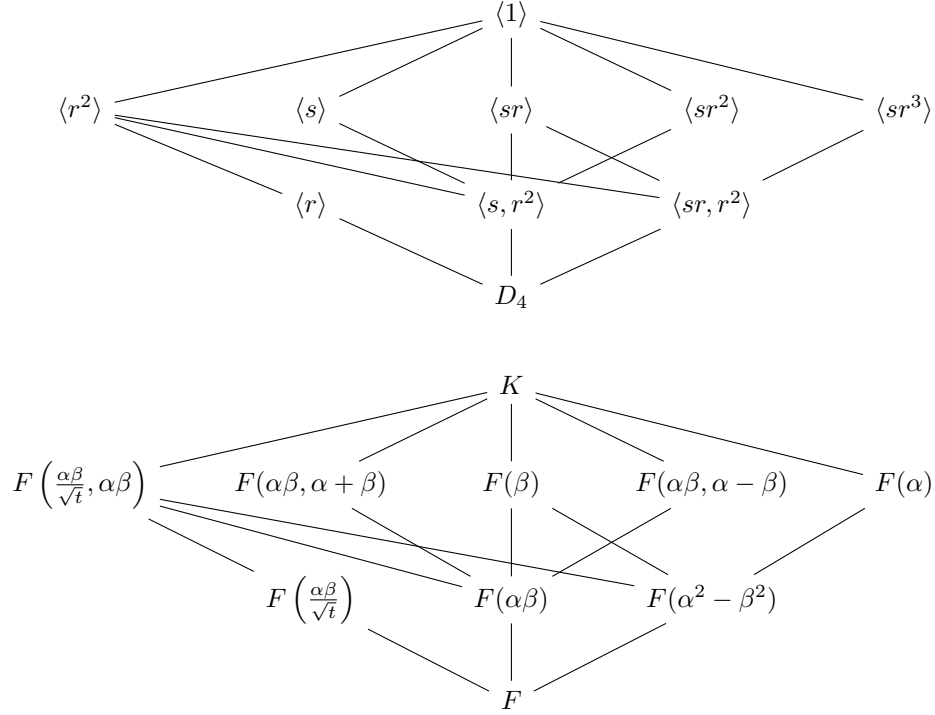
$\alpha\beta = \sqrt{r^2 - t}$ is in $K \iff \alpha\beta$ is fixed by $\text{Gal}(K/F)$. Suppose, $\forall g \in G(K/F)$, $g(\alpha) = -\alpha \implies g(\beta) = g(\alpha\beta)/g(\alpha) = \alpha\beta / -\alpha = -\beta$. g^2 takes $\alpha \rightarrow \alpha$ and $\beta \rightarrow \beta$. Thus g has order 2. Thus $g = \sigma$.

If $\forall g \in G(K/F)$, $g(\alpha) = \beta \implies g(\beta) = g(\alpha\beta)/g(\alpha) = \alpha\beta / \beta = \alpha$. g^2 takes $\alpha \rightarrow \beta \rightarrow \alpha$ and $\beta \rightarrow \alpha \rightarrow \beta$. Thus g has order 2 and $g = \rho$.

If $\forall g \in G(K/F)$, $g(\alpha) = -\beta$. Then $g(\beta) = g(\alpha\beta)/g(\alpha) = \alpha\beta / -\beta = -\alpha$. $g = \rho \circ \sigma$ because $\alpha \rightarrow -\alpha \rightarrow -\beta = g(\alpha)$ and $\beta \rightarrow -\beta \rightarrow -\alpha = g(\beta)$. And $g = \sigma \circ \rho$ because $\alpha \rightarrow \beta \rightarrow -\beta = g(\alpha)$ and $\beta \rightarrow \alpha \rightarrow -\alpha = g(\beta)$.

Thus $\text{Gal}(K/F)$ is D_2 .

$G(K/F) = C_4$ if $G(K/F) \neq D_4, C_4$ so if $r^2 - t$ is square in $F[\sqrt{t}]$ and $r^2 - t$ is not square in F . If $r^2 - t$ is square in $F[\sqrt{t}]$, $\exists x, y \in F[\sqrt{t}]$, $r^2 - t = (x + y\sqrt{t})^2 = x^2 + 2xy\sqrt{t} + y^2$ since $r^2 - t \in K$, $2xy\sqrt{t} = 0$. If $y = 0$, then $r^2 - t = x^2 \implies r^2 - t$ is a square in K which is a contradiction. Thus $x = 0$. Thus $r^2 - t = y^2t \implies y^2 = \frac{r^2 - t}{t}$. Thus $G(K/F) = C_4 \iff \frac{r^2 - t}{t}$ square in F .



Since all the subgroups with lines between them have indexes of 2, we know the degree of any two subfields with a line between them have a degree of 2, so once we find one element which is fixed not in the below subfields, we know what the extension of the subfield above it is. Note I worked with Prajith to make the diagrams. r is a rotation counter clockwise and s is a vertical reflection across the y axis.

For $\langle r \rangle$ and $\langle s, r^2 \rangle$ we used the results above to help us.

$F(\alpha\beta)/F$. Thus $f = \alpha\beta + \sqrt{t} = \alpha\beta + \sqrt{t} = \alpha\beta + \alpha^2 - r$

$\langle r^2 \rangle \triangleleft D_4 \implies \text{Gal}(F(\sqrt{t}, \alpha\beta)/F) = D_4/\mu_2 = \{r, s, rs, 1\}$

$r(f) = -\alpha\beta + \beta^2 - r$

$r(r(f)) = \alpha\beta + \alpha^2 - r$

$s(f) = \alpha\beta + \beta^2 - r$ $s(s(f)) = \alpha\beta + \alpha^2 - r$

$s(r(f)) = -\alpha\beta + \alpha^2 - r$ $s(r(s(r(f)))) = s(r(-\alpha\beta + \alpha^2 - r)) = \alpha\beta + \alpha^2 - r$

Thus f has a orbit of size 4. Thus $F(\alpha\beta, \sqrt{t}) = F(\alpha\beta + \sqrt{t})$

□